Task 1: Define Your Research Questions

1. Primary Research Problem

How do advertised Airbnb listing prices vary across different cities, room types and accommodation characteristics?

The purpose of this research is to investigate differences in Airbnb advertised prices across ten selected cities. The analysis will explore how location, room type, property characteristics and accommodation capacity relate to advertised prices. The findings will help identify patterns and differences in the Airbnb accommodation market.

2. Supporting Research Questions

Research Question 1: Price differences across room types and cities

How do advertised Airbnb listing prices differ between Entire home/apartment, Private room and other room types within individual cities?

This question will help determine whether certain room types are generally advertised at higher prices and whether these differences vary between cities.

Research Question 2: Accommodation characteristics and advertised prices

Are the number of bedrooms and the accommodation capacity (accommodates) associated with differences in advertised Airbnb listing prices?

This question will investigate whether properties with more bedrooms or the ability to accommodate more guests tend to have higher advertised prices.

Research Question 3: Host performance indicators across cities

Do host response indicators, such as host response time, response rate and superhost status, differ across the selected cities?

This question will help identify differences in host characteristics and service-related indicators across Airbnb markets.

Research Question 4: Review activity and listing-level ratings

Is the amount of review activity associated with listing-level review ratings?

This question will investigate whether listings with more recorded reviews tend to have different review scores from listings with fewer reviews. Review counts will be calculated from the

Reviews dataset, while listing-level ratings will be obtained from the Listings dataset where reliable matching records are available.

3. Research Objectives

The research will aim to:

1. Compare advertised Airbnb prices across the ten selected cities.
2. Examine price differences between room types within individual cities.
3. Investigate the relationship between accommodation capacity, number of bedrooms and advertised prices.
4. Compare host response indicators across cities.
5. Examine historical review activity and investigate its relationship with listing-level review ratings where the available data supports a reliable comparison.

4. Expected Research Outcomes

The research is expected to identify differences in advertised prices across cities and room types, highlight accommodation characteristics associated with pricing, and describe variations in host performance indicators and review activity.

The findings will support a better understanding of the Airbnb accommodation market and provide evidence-based insights into listing characteristics and market differences.